

Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Static · 30 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Static · 30 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Static · 30 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/5_Georgia/Static · 30 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Static · 30 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/7_Henry/Static · 30 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static · 31 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/9_Joselyn/Static · 30 trials
Training data path 9	/home/jestin/ThesisRepo/ML/NewTestData/10_Josh/Static · 30 trials
Training data path 10	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Static · 31 trials
Training data path 11	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Static · 30 trials
Training data path 12	/home/jestin/ThesisRepo/ML/NewTestData/13_Nat/Static · 32 trials
Training data path 13	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Static · 30 trials
Training data path 14	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Static · 0 trials
Training data path 15	/home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Static · 30 trials
Training data path 16	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Static · 30 trials
Total training paths	16 · 454 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Static
Report output dir	/home/jestin/ThesisRepo/ML/saved_reports/AugmentationExperiments

Sensor selection

Hands	left
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	88 channels
Classes	6: Double_Closed_Fist, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	off · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats+fft → feature dim 4752
Test size · seed · CV folds	0.2 · random_state=42 · 10-fold stratified
Sample counts	train (post-aug)=726 · test=91

Augmentation (training only)

Enabled	yes
Copies per train trial	1
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.9, 1.1)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (10-fold)	CV std	Hold-out test acc
Stacking (SVM+RF+KNN+LogReg)	0.9945	0.0110	1.0000
Voting Soft (SVM+RF+KNN+LogReg)	0.9918	0.0175	0.9670
Random Forest	0.9959	0.0088	0.9560

Per-class hold-out F1

Class	Stacking	Voting (soft)	Random Forest	Support
Double_Closed_Fist	1.000	0.970	0.941	16
Double_Okay	1.000	0.966	0.968	15
Double_Open_Palm	1.000	0.968	0.966	15
Double_Phone	1.000	0.933	0.933	15
Double_Pistol	1.000	0.966	0.929	15
Double_Spiderman	1.000	1.000	1.000	15
macro avg	1.000	0.967	0.956	91

External test (NewTestData)

Algorithm	External test acc
Random Forest	0.8333
Stacking (SVM+RF+KNN+LogReg)	0.8333

Voting Soft (SVM+RF+KNN+LogReg)	0.7667
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Per-class external accuracy

Class	Random Forest	Stacking	Voting (soft)	Support
Double_Closed_Fist	1.000	1.000	0.800	5
Double_Okay	1.000	1.000	0.800	5
Double_Open_Palm	1.000	1.000	1.000	5
Double_Phone	0.000	0.000	0.000	5
Double_Pistol	1.000	1.000	1.000	5
Double_Spiderman	1.000	1.000	1.000	5